

WEEK 12 REPORT

We successfully saved our model using Pickle, naming it “stock_direction_model.pkl”.

The following section will discuss environment dependencies including OS, Python version, Python packages used, and versions:

Operating System

- OS: Linux-based JupyterHub environment

Python Version

- Python Version: 3.12.6

Python Packages and Libraries Used

The following libraries were used in the development of this project:

- Core Data Processing
 - Pandas
 - Numpy
 - OS
 - re
 - gc
 - Collections
- Visualization
 - Matplotlib
 - Seaborn
- Time Series Analysis
 - Statsmodels
- Machine Learning
 - Scikit-learn
 - LogisticRegression
 - SVC
 - StandardScaler
 - PCA
 - GridSearchCV / RandomizedSearchCV
 - StratifiedKFold
 - Metrics (accuracy, F1-score, confusion matrix, classification report)
- Imbalanced Data Handling
 - Imbalanced-learn

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- SMOTE
- Pipeline
- Text Processing
 - scikit-learn (TF-IDF, stop words)
- Gradient Boosting Model
 - XGboost
 - XGBClassifier
- Natural Language Processing (LLM / Transformer)
 - Transformers
 - AutoTokenizer
 - AutoModelForSequenceClassification
 - Torch
- Model Explainability
 - Shap
- Model Persistence
 - joblib

This model will be deployed in batch mode on a daily schedule over real-time inference. Since the model uses data aggregated on a daily level, the model does not need to make an inference more than once a day. Batch mode deployed on a daily schedule is a perfect candidate because it will generate predictions based on the previous day's closing stock price and news sentiment and deliver to the client on a daily cadence. It can do this without racking up high costs like a real-time inference deployment would, and it uses far fewer computational resources.

Two model performance metrics that should be monitored in production are accuracy and macro average F1 score. These performance metrics were used for model evaluation, so it is important they are both consistently monitored to ensure they do not drift too far from validation and test set performance. If they do, this indicates that there might be data drift present, and the model likely needs to be retrained.

Another model performance metric to monitor is the decline in loss that each feature contributes. We used this as a metric in previous weeks to evaluate which predictors are the most important in contributing to model performance, and this could be a good metric to ensure there is limited concept drift. If the decline in loss drops too low for any given feature, this could indicate that these predictors are no longer relevant for predicting stock price direction and can be removed or substituted for others.

A business metric that should be tracked in production is revenue generated. In the first week, we made an initial assumption of a \$1MM investment, and correct predictions of this model could allow the client to grow that investment by buying stock at relatively low prices and selling stock at relatively high prices.

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Generating substantial revenue could justify the use of a machine learning model over headcount for a human day trader, and allow the business to grow while keeping expenses limited.

The final metric to monitor is latency. When a client asks for a stock price direction prediction, the decision needs to be returned in a reasonable amount of time. If the client is left waiting too long, user satisfaction will likely decrease, and we risk a higher abandonment rate. Infrastructure and system architecture may also affect latency, but model complexity can also play a role in high response times. Low latency will likely lead to a happier user experience, and a higher chance for that user to return to the platform.

When establishing our thresholds for green, yellow, and red flags, it is important to keep in mind that stock price prediction is particularly difficult. Even if our models have an accuracy or F1 score slightly higher than random guessing, it could be profitable in the real world.

Our green threshold will be if the model's accuracy and/or F1 score is higher than 0.53. Our model's performance on the training data indicates that an accuracy of around 0.53 is performing well. This was around how well our model performed on the validation and test data.

An accuracy and/or F1 score between 0.51 and 0.52 will be seen as our yellow threshold. This indicates that our model is still exceeding random guessing, but not by much. This is an indication that we need to closely monitor our models' performance and take action if these metrics drop any lower.

Our red threshold will be when our models accuracy and/or F1 score drops to 0.5 or below. This means our model is performing at or below the level of random guessing. At this point, the model has almost no value and should be pulled out of production.

If our model is within the green threshold, it will remain in production with our standard monitoring and routine retraining. If it falls within the yellow threshold, the model will also remain in production but will require more monitoring. We would also have an analyst examine possible data or concept drift. If the model falls into the red threshold it would be pulled from production. Similar to how we built our initial model, we could use this time to rebuild a new model using what we know about data or concept drift.

Our model would need to be trained regularly instead of kept static. Both stock prices and news sentiment are constantly changing. We would need to update our model with new data as frequently as possible in order to avoid concept drift, when historical patterns become less important in the future. Since companies like Apple report their earnings every quarter, it is logical to retrain our model each quarter with updated stock price and new sentiment data. This means we will retrain the model every three months. Combined with monitoring our model's performance, frequently retraining our model with new data will help it retain its relevance and robustness over time.

Data drift occurs when the distribution of predictor variables changes in relation to the training data over time. This change can lead to a drop in model performance. In our case, this would most likely be the distribution of sentiment polarity (our main predictor) changing over time. If a major shift in Apple news sentiment happens in the future, it could completely change the distribution and decrease model

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performance. While frequently retraining our model would help mitigate this problem, we could also make sure to monitor the distributions of all of our predictor variables to ensure there is not a major change.

Concept drift occurs when the relationship between predictor variables and target variables changes over time. In our case, this could be if new sentiment becomes completely disconnected from stock change direction. This could lead to a drop in our model's performance. In order to mitigate this risk, retraining the model will be highly beneficial. If our model is retrained every quarter, it will factor in any potential changes between new sentiment and stock price direction.